

Name: _____

Date: _____

Day 1

Period: _____

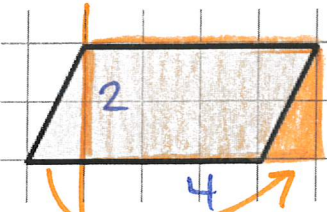
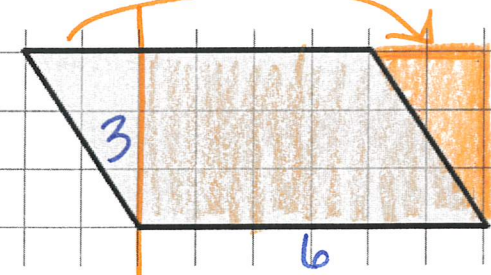
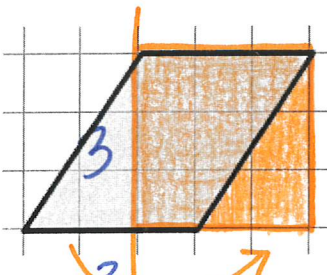
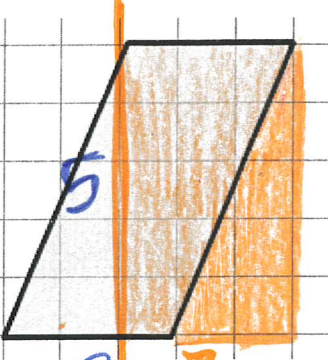
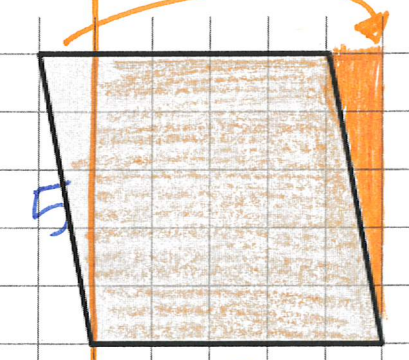
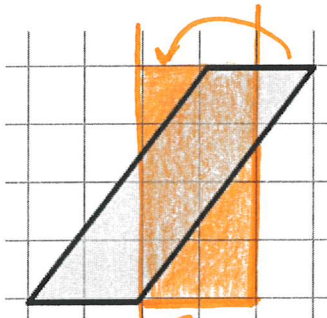
Decomposing and Rearranging Parallelograms

Cut the area of the SQUARE in HALF on the diagonal.

Original shape	Decomposed shape	Describe how to find the area
 24 units² 4 units 6 units	 $A = bh$ $A = 6(4) = 24 \text{ units}^2$ 4 6	Multiply the <u>base</u> and <u>height</u> of the parallelogram.

Formula for finding area of a parallelogram: $A = bh$

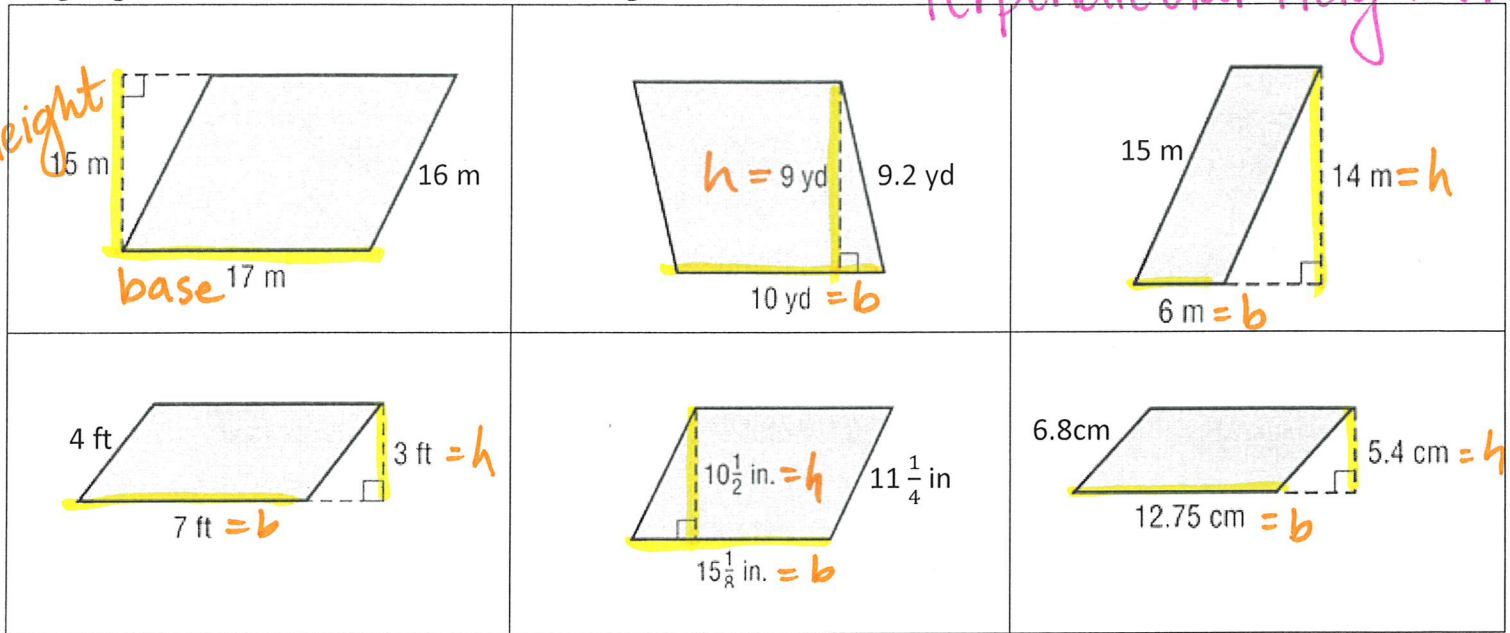
Cut the area of the square or rectangle in half on the diagonal. Find the area of the triangle.

 $A = bh$ $A = 4(2) = 8 \text{ units}^2$	 $A = bh$ $A = 6(3) = 18 \text{ u}^2$	 $A = bh$ $A = 3(3) = 9 \text{ u}^2$
 $A = bh$ $A = 3(5)$ $A = 15 \text{ units}^2$	 $A = bh$ $A = 5(5) = 25 \text{ units}^2$	 $A = bh$ $A = 2(4) = 8 \text{ units}^2$

Area of Parallelograms

Highlight and label the base and the height

Perpendicular Height (90°)



Answer the following questions

1. Manu is a sunflower farmer. His farm is in the shape of a parallelogram with a height measuring 3.5 kilometers and a base measuring 4.25 kilometers. What is the total land area Manu uses?

Base: 4.25 km

Height: 3.5 km

Formula: $A = bh$

Substitute: $A = 4.25(3.5)$

$$A = 14.875 \text{ km}^2$$



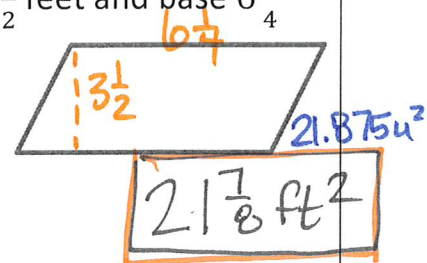
2. Tara and Veronica are in charge of making a banner for the volleyball game this Saturday. How much poster paper will they need for a parallelogram-shaped banner with height $3\frac{1}{2}$ feet and base $6\frac{1}{4}$ feet?

Base: $6\frac{1}{4}$

Height: $3\frac{1}{2}$

Formula: $A = bh$

Substitute: $A = 6\frac{1}{4}(3\frac{1}{2}) = \frac{25}{4} \cdot \frac{7}{2} = \frac{175}{8}$



3. Joseph is painting the flag of Brunei (a country in Southeast Asia) for a geography project at school. How many square inches will he cover with white paint?

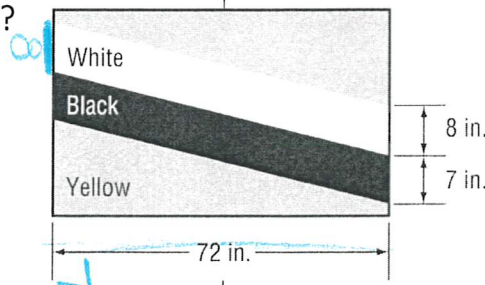
Base: 8 in

Height: 72 in

Formula: $A = bh$

Substitute: $A = 8(72)$

$$A = 576 \text{ in}^2$$



4. Use the flag from Exercise 3. How many square inches will Joseph cover with black paint?

Base: 7

Height: 72

Formula: $A = bh$

Substitute: $A = 7(72)$

$$A = 504 \text{ in}^2$$

$$\begin{array}{r} 72 \\ \times 7 \\ \hline 504 \end{array}$$